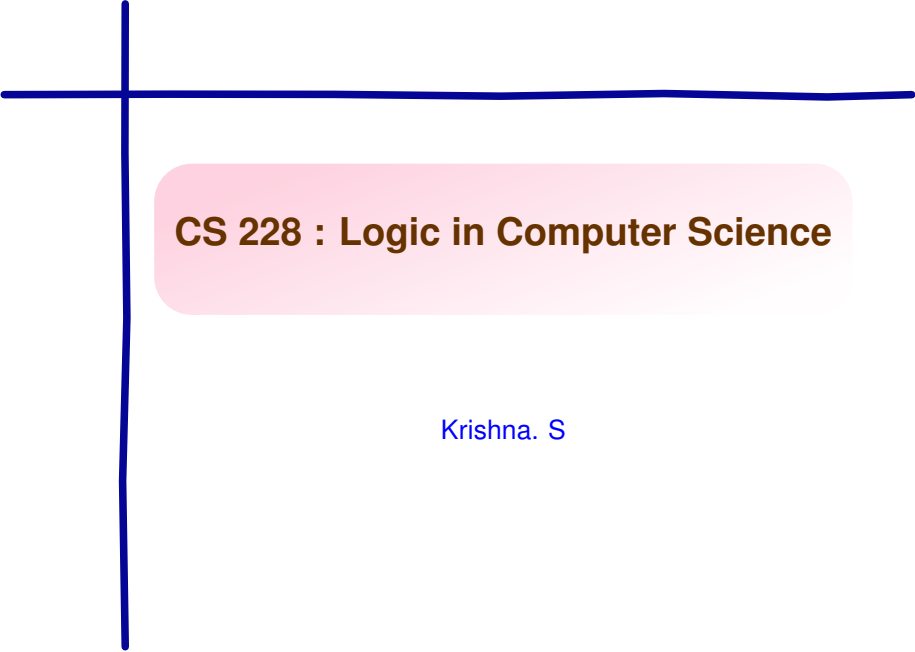




CS 228 : Logic in Computer Science

Krishna. S

So Far

- ▶ Next operator to GNBA
- ▶ Recall closure of a formula, elementary sets

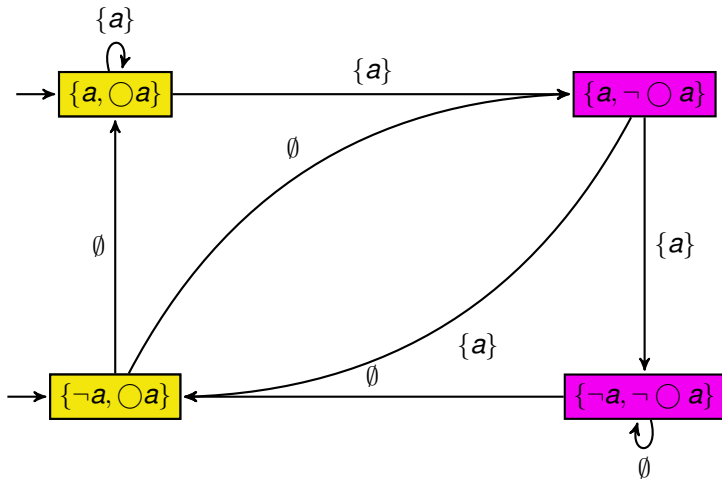
LTL φ to GNBA G_φ

- ▶ States of G_φ are elementary sets B_i
- ▶ For a word $w = A_0A_1A_2\dots$ the sequence of states $\sigma = B_0B_1B_2\dots$ will be a run for w
- ▶ σ will be accepting iff $w \models \varphi$ iff $\varphi \in B_0$
- ▶ In general, a run $B_iB_{i+1}\dots$ for $A_iA_{i+1}\dots$ is accepting iff $A_iA_{i+1}\dots \models \psi$ for all $\psi \in B_i$.

LTL to GNBA

- ▶ Let $\varphi = \bigcirc a$.
- ▶ Subformulae of φ : $\{a, \bigcirc a\}$. Let $A = \{a, \bigcirc a, \neg a, \neg \bigcirc a\}$.
- ▶ Possibilities at each state
 - ▶ $\{a, \bigcirc a\}$
 - ▶ $\{\neg a, \bigcirc a\}$
 - ▶ $\{a, \neg \bigcirc a\}$
 - ▶ $\{\neg a, \neg \bigcirc a\}$
- ▶ Our initial state(s) must guarantee truth of $\bigcirc a$. Thus, initial states: $\{a, \bigcirc a\}$ and $\{\neg a, \bigcirc a\}$

LTL to GNBA



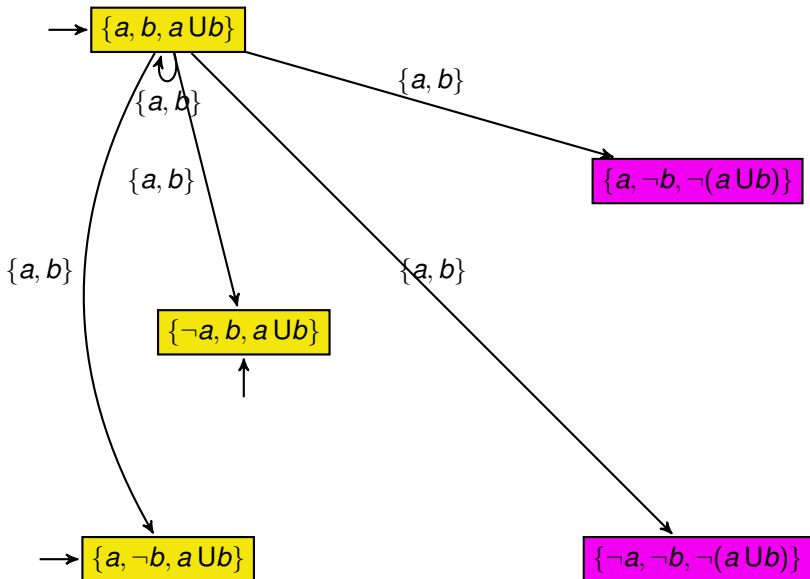
LTL to GNBA

- ▶ Claim : Runs from a state labelled set B indeed satisfy B
- ▶ No good states. All words having a run from a start state are accepted.
- ▶ Correct by construction!
- ▶ Automaton for $\neg \bigcirc a$ same, except for the start states.

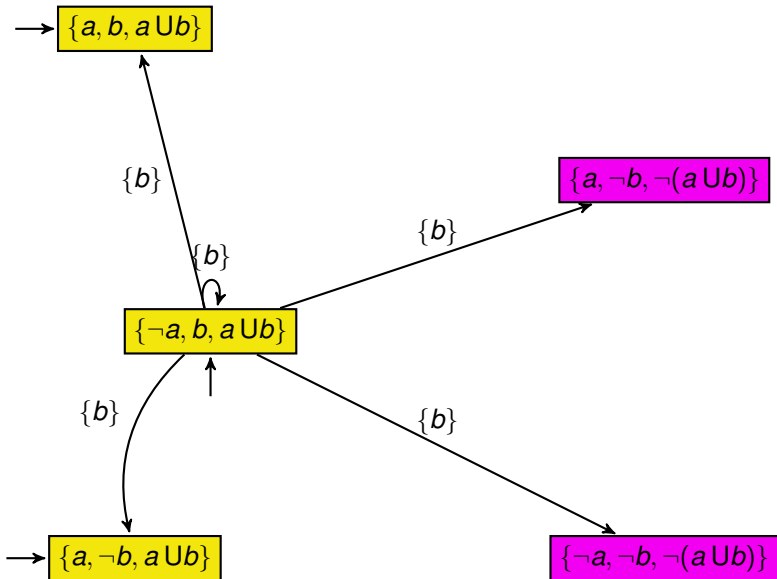
LTL to GNBA

- ▶ Let $\varphi = a \text{ U } b$.
- ▶ Subformulae of $\varphi : \{a, b, a \text{ U } b\}$,
 $cl(\varphi) = \{a, \neg a, b, \neg b, a \text{ U } b, \neg(a \text{ U } b)\}$.
- ▶ Possibilities at each state
 - ▶ $\{a, \neg b, a \text{ U } b\}$
 - ▶ $\{\neg a, b, a \text{ U } b\}$
 - ▶ $\{a, b, a \text{ U } b\}$
 - ▶ $\{a, \neg b, \neg(a \text{ U } b)\}$
 - ▶ $\{\neg a, \neg b, \neg(a \text{ U } b)\}$
- ▶ Our initial state(s) must guarantee truth of $a \text{ U } b$. Thus, initial states: $\{a, b, a \text{ U } b\}$ and $\{\neg a, b, a \text{ U } b\}$ and $\{a, \neg b, a \text{ U } b\}$.

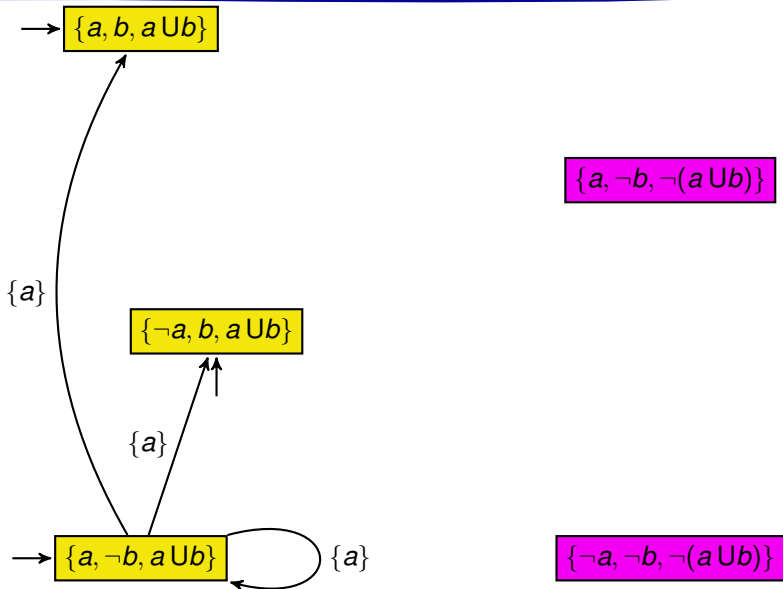
LTL to GNBA



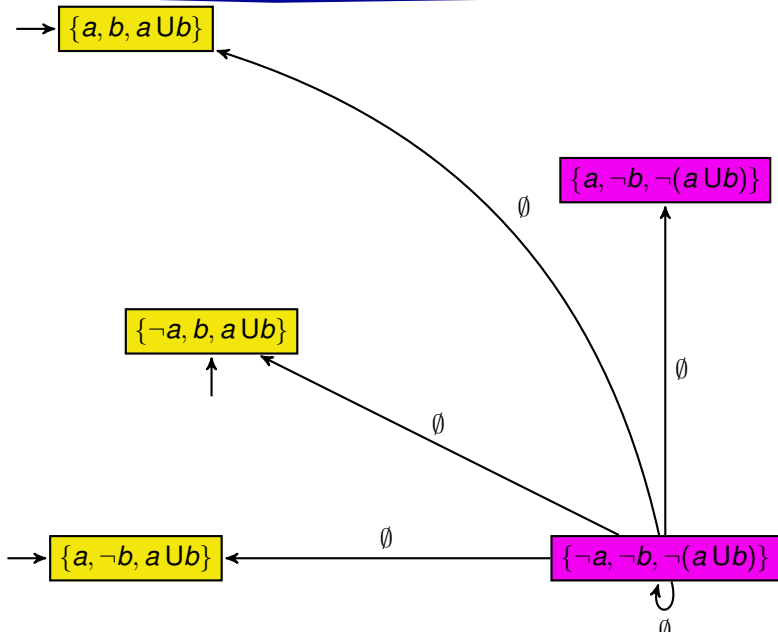
LTL to GNBA



LTL to GNBA



LTL to GNBA



LTL to GNBA

→ $\{a, b, a \cup b\}$

$\{\neg a, b, a \cup b\}$



→ $\{a, \neg b, a \cup b\}$

$\{a\}$



$\{a, \neg b, \neg(a \cup b)\}$

$\{a\}$



$\{\neg a, \neg b, \neg(a \cup b)\}$

LTL to GNBA : Accepting States

→ $\{a, b, a \cup b\}$

$\{a, \neg b, \neg(a \cup b)\}$

$\{\neg a, b, a \cup b\}$



→ $\{a, \neg b, a \cup b\}$

$\{\neg a, \neg b, \neg(a \cup b)\}$

LTL to GNBA

Construct GNBA for $\neg(a \text{ U } b)$.

LTL to GNBA

- ▶ Let $\varphi = a \text{ U } (\neg a \text{ U } c)$. Let $\psi = \neg a \text{ U } c$
- ▶ Subformulae of $\varphi : \{a, \neg a, c, \psi, \varphi\}$.
 $cl(\varphi) = \{a, \neg a, c, \neg c, \psi, \neg\psi, \varphi, \neg\varphi\}$.
- ▶ Possibilities at each state
 - ▶ $\{a, c, \psi, \varphi\}$
 - ▶ $\{\neg a, c, \psi, \varphi\}$
 - ▶ $\{a, \neg c, \neg\psi, \varphi\}$
 - ▶ $\{a, \neg c, \neg\psi, \neg\varphi\}$
 - ▶ $\{\neg a, \neg c, \psi, \varphi\}$
 - ▶ $\{\neg a, \neg c, \neg\psi, \neg\varphi\}$

LTL to GNBA

→ $\{a, c, \psi, \varphi\}$

$\{\neg a, \neg c, \psi, \varphi\}$ ←

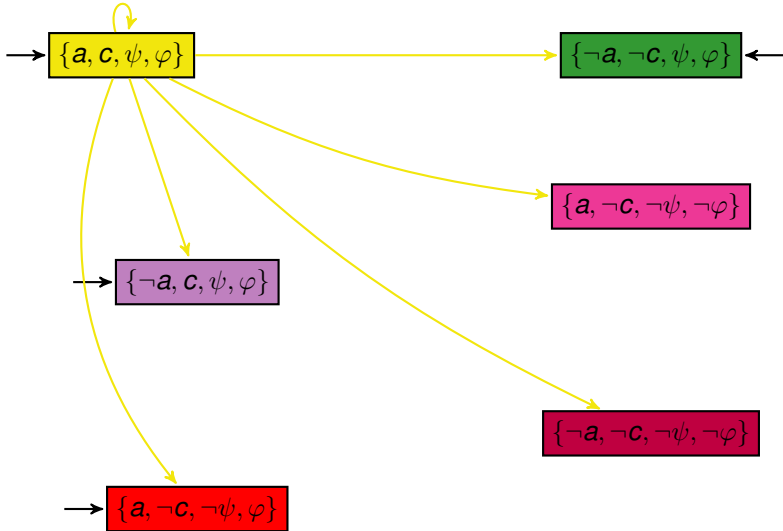
→ $\{\neg a, c, \psi, \varphi\}$

$\{a, \neg c, \neg \psi, \neg \varphi\}$

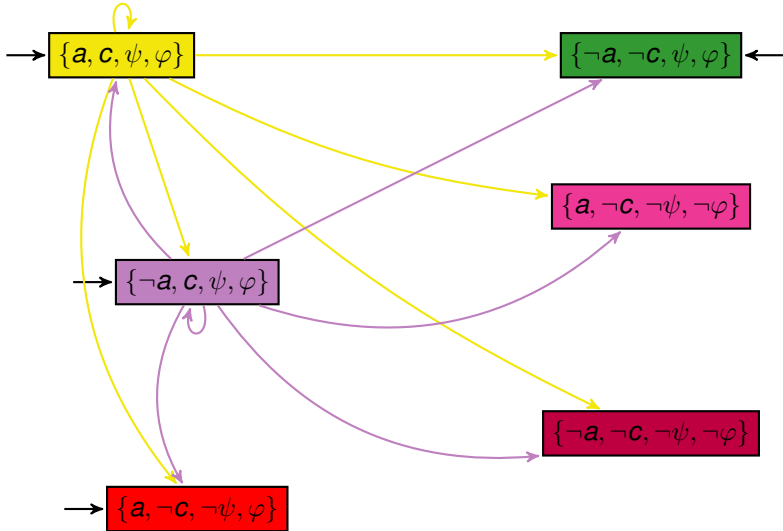
→ $\{a, \neg c, \neg \psi, \varphi\}$

$\{\neg a, \neg c, \neg \psi, \neg \varphi\}$

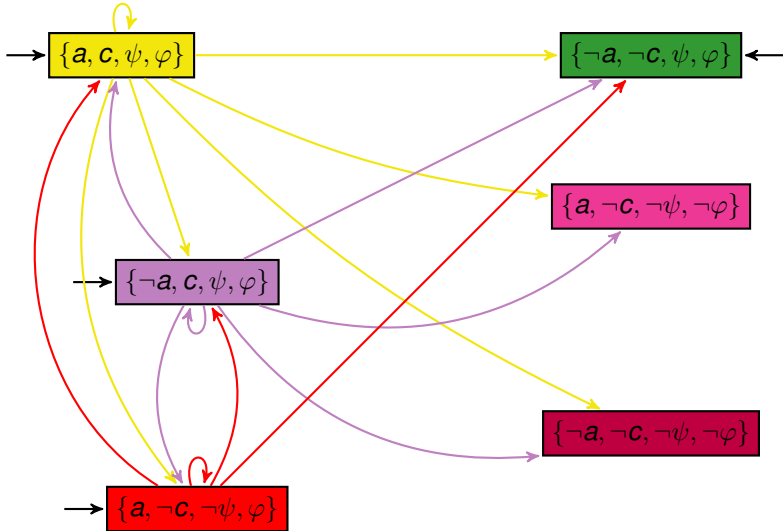
LTL to GNBA



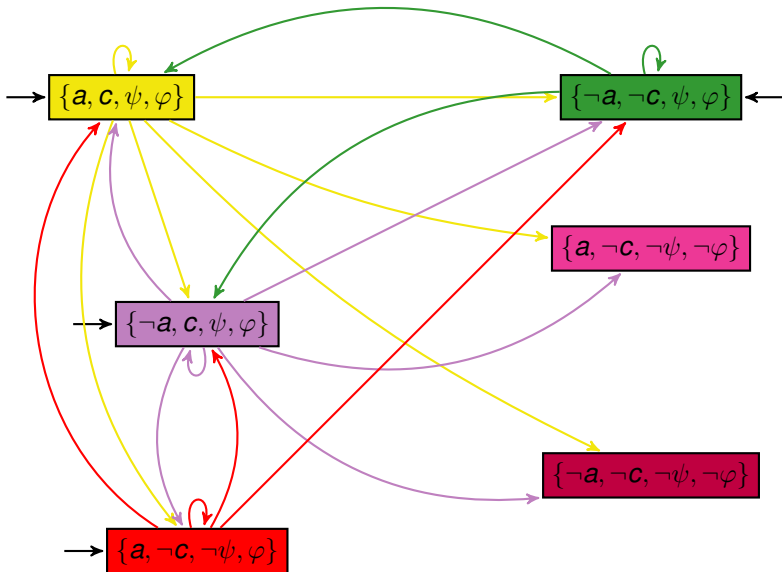
LTL to GNBA



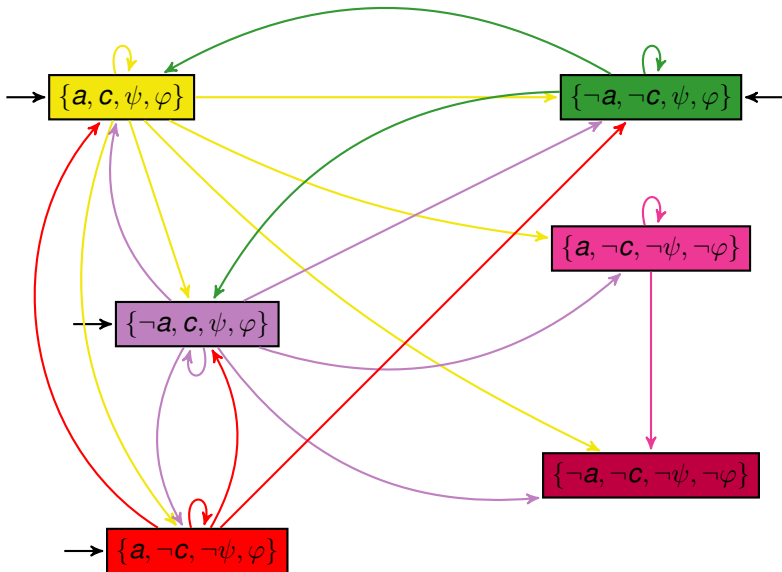
LTL to GNBA



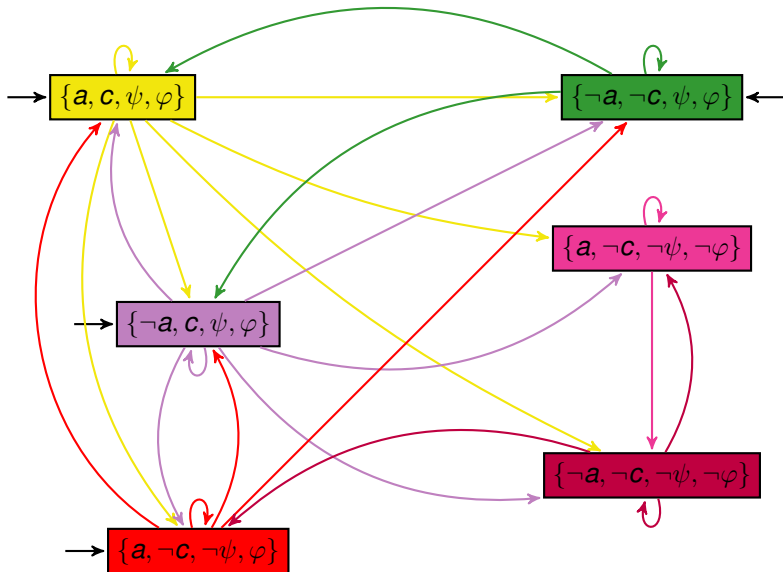
LTL to GNBA



LTL to GNBA



LTL to GNBA



GNBA Acceptance Condition

- ▶ $\psi = \neg a \mathbf{U} c$
- ▶ $\varphi = a \mathbf{U} \psi$
- ▶ $F_1 = \{B \mid \psi \in B \rightarrow c \in B\}$
- ▶ $F_2 = \{B \mid \varphi \in B \rightarrow \psi \in B\}$
- ▶ $\mathcal{F} = \{F_1, F_2\}$

Final States

$$\rightarrow \{a, c, \psi, \varphi\} \in F_1, F_2$$

$$\{\neg a, \neg c, \psi, \varphi\} \in F_2 \leftarrow$$

$$\{a, \neg c, \neg \psi, \neg \varphi\} \in F_1, F_2$$

$$\rightarrow \{\neg a, c, \psi, \varphi\} \in F_1, F_2$$

$$\{\neg a, \neg c, \neg \psi, \neg \varphi\} \in F_1, F_2$$

$$\rightarrow \{a, \neg c, \neg \psi, \varphi\} \in F_1$$

Putting Together

- ▶ Given φ , build $CI(\varphi)$, the set of all subformulae of φ and their negations

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Putting Together

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- ▶ Consider those $B \subseteq CI(\varphi)$ which are **consistent**
 - ▶ $\varphi_1 \wedge \varphi_2 \in B \leftrightarrow \varphi_1 \in B \text{ and } \varphi_2 \in B$
 - ▶ $\psi \in B \rightarrow \neg\psi \notin B$ and $\psi \notin B \rightarrow \neg\psi \in B$
 - ▶ Whenever $\psi_1 \cup \psi_2 \in CI(\varphi)$,
 - ▶ $\psi_2 \in B \rightarrow \psi_1 \cup \psi_2 \in B$
 - ▶ $\psi_1 \cup \psi_2 \in B$ and $\psi_2 \notin B \rightarrow \psi_1 \in B$

Putting Together

Given φ over AP , construct $A_\varphi = (Q, 2^{AP}, \delta, Q_0, \mathcal{F})$,

- ▶ $Q = \{B \mid B \subseteq Cl(\varphi) \text{ is consistent} \}$
- ▶ $Q_0 = \{B \mid \varphi \in B\}$
- ▶ $\delta : Q \times 2^{AP} \rightarrow 2^Q$ is such that

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 - ▶ For $C = B \cap AP$, $\delta(B, C)$ is enabled and is defined as :
 - ▶ If $\bigcirc\psi \in Cl(\varphi)$, $\bigcirc\psi \in B$ iff $\psi \in \delta(B, C)$
 - ▶ If $\varphi_1 \cup \varphi_2 \in Cl(\varphi)$,
 $\varphi_1 \cup \varphi_2 \in B$ iff $(\varphi_2 \in B \vee (\varphi_1 \in B \wedge \varphi_1 \cup \varphi_2 \in \delta(B, C)))$

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- ▶ $\mathcal{F} = \{F_{\varphi_1 \mathbf{U} \varphi_2} \mid \varphi_1 \mathbf{U} \varphi_2 \in Cl(\varphi)\}$, with
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- ▶ $\mathcal{F} = \{F_{\varphi_1 \mathbf{U} \varphi_2} \mid \varphi_1 \mathbf{U} \varphi_2 \in Cl(\varphi)\}$, with
 $F_{\varphi_1 \mathbf{U} \varphi_2} = \{B \in Q \mid \varphi_1 \mathbf{U} \varphi_2 \in B \rightarrow \varphi_2 \in B\}$
- ▶ Prove that $L(\varphi) = L(A_\varphi)$

$$L(\varphi) \subseteq L(A_\varphi)$$

Let $\sigma = A_0A_1A_2\cdots \in L(\varphi)$. Show that there is an accepting run $B_0A_0B_1A_1B_2A_2\cdots$ in A_φ for σ , B_i are the states, such that $B_i = \{\psi \mid A_iA_{i+1}\cdots \models \psi\}$.

Structural induction on φ

- ▶ $\varphi = a$. All starting states contain a , and can go to all successor states with all combinations of propositions.
- ▶ If $a \in B_i$, every run starting at B_i starts with a . Hence, $A_iA_{i+1}\cdots \models a$

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- ▶ $\varphi = \bigcirc a$, then all initial states contain $\bigcirc a$, and all successor states contain a by construction.
- ▶ If $\bigcirc a \in B_i$, then by construction, $B_{i+1} \in \delta(B_i, B_i \cap AP)$ iff $a \in B_{i+1}$, for every successor B_{i+1} .
Then $A_{i+1}\cdots \models a$, and hence $A_iA_{i+1}\cdots \models \bigcirc a$.

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- If $\varphi_1 \cup \varphi_2 \in B_i$, then by construction, either $\varphi_2 \in B_i$ or $\varphi_1, \varphi_1 \cup \varphi_2 \in B_i$. If $\varphi_2 \in B_i$ then $A_iA_{i+1}\cdots \models \varphi_2$ by induction hypothesis, and hence, $A_iA_{i+1}\cdots \models \varphi_1 \cup \varphi_2$.

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- ▶ If $\varphi_2 \in B_k$ for $k > i$, and $\varphi_1, \varphi_1 \cup \varphi_2 \in B_{i+1}, \dots, B_{k-1}$ then $A_iA_{i+1}\cdots \models \varphi_1 \cup \varphi_2$.
- ▶ When is $B_iB_{i+1}B_{i+2}\cdots$ an accepting run?
- ▶ $B_j \in F_{\varphi_1 \cup \varphi_2}$ for infinitely many $j \geq i$.

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Let $\sigma = A_0A_1A_2 \dots \in L(\varphi)$. Show that there is an accepting run $B_0A_0B_1A_1B_2A_2 \dots$ in A_φ for σ , B_i are the states, such that $B_i = \{\psi \mid A_iA_{i+1} \dots \models \psi\}$.

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- ▶ When is $B_iB_{i+1}B_{i+2} \dots$ an accepting run?
- ▶ $B_j \in F_{\varphi_1 \cup \varphi_2}$ for infinitely many $j \geq i$.
- ▶ $\varphi_2 \notin B_j$ or $\varphi_2, \varphi_1 \cup \varphi_2 \in B_j$ for infinitely many $j \geq i$.
- ▶ By construction, there is an accepting run where $\varphi_2 \in B_k$ for some $k \geq i$. Hence, $A_iA_{i+1} \dots \models \varphi_1 \cup \varphi_2$.

$$L(\varphi) \subseteq L(A_\varphi)$$

Let $\sigma = A_0A_1A_2\cdots \in L(\varphi)$. Show that there is an accepting run $B_0A_0B_1A_1B_2A_2\cdots$ in A_φ for σ , B_i are the states, such that $B_i = \{\psi \mid A_iA_{i+1}\cdots \models \psi\}$.

- If $\neg(\varphi_1 \cup \varphi_2) \in B_i$, then either $\neg\varphi_1, \neg\varphi_2 \in B_i$ or $\varphi_1, \neg\varphi_2 \in B_i$. If $\neg\varphi_1, \neg\varphi_2 \in B_i$ then $A_iA_{i+1}\cdots \models \neg(\varphi_1 \cup \varphi_2)$.

$$L(\varphi) \subseteq L(A_\varphi)$$

Let $\sigma = A_0A_1A_2\cdots \in L(\varphi)$. Show that there is an accepting run $B_0A_0B_1A_1B_2A_2\cdots$ in A_φ for σ , B_i are the states, such that $B_i = \{\psi \mid A_iA_{i+1}\cdots \models \psi\}$.

- ▶ If $\neg(\varphi_1 \cup \varphi_2) \in B_i$, then either $\neg\varphi_1, \neg\varphi_2 \in B_i$ or $\varphi_1, \neg\varphi_2 \in B_i$. If $\neg\varphi_1, \neg\varphi_2 \in B_i$ then $A_iA_{i+1}\cdots \models \neg(\varphi_1 \cup \varphi_2)$.
- ▶ If $\varphi_1, \neg\varphi_2 \in B_i$, then by construction, $B_{i+1} \in \delta(B_i, B_i \cap AP)$ iff $\varphi_1, \neg\varphi_2 \in B_{i+1}$ or $\neg\varphi_1, \neg\varphi_2 \in B_{i+1}$.

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- ▶ If $\varphi_1, \neg\varphi_2 \in B_i$, then by construction, $B_{i+1} \in \delta(B_i, B_i \cap AP)$ iff $\varphi_1, \neg\varphi_2 \in B_{i+1}$ or $\neg\varphi_1, \neg\varphi_2 \in B_{i+1}$.
- ▶ Either case, $A_iA_{i+1}\cdots \models \neg(\varphi_1 \cup \varphi_2)$

$$L(A_\varphi) \subseteq L(\varphi)$$

For a sequence $B_0 B_1 B_2 \dots$ of states satisfying

- ▶ $B_{i+1} \in \delta(B_i, A_i)$,
- ▶ $\forall F \in \mathcal{F}, B_j \in F$ for infinitely many j ,

we have $\psi \in B_0 \leftrightarrow A_0 A_1 \dots \models \psi$

- ▶ Structural Induction on ψ . Interesting case : $\psi = \varphi_1 \mathbf{U} \varphi_2$
- ▶ Assume $A_0 A_1 \dots \models \varphi_1 \mathbf{U} \varphi_2$. Then $\exists j \geq 0, A_j A_{j+1} \dots \models \varphi_2$ and $A_i A_{i+1} \dots \models \varphi_1, \varphi_1 \mathbf{U} \varphi_2$ for all $i \leq j$.
- ▶ By induction hypothesis (applied to φ_1, φ_2), we obtain $\varphi_2 \in B_j$ and $\varphi_1 \in B_i$ for all $i \leq j$
- ▶ By induction on j , $\varphi_1 \mathbf{U} \varphi_2 \in B_j, \dots, B_0$.

$$L(A_\varphi) \subseteq L(\varphi)$$

For a sequence $B_0 B_1 B_2 \dots$ of states satisfying

- (a) $B_{i+1} \in \delta(B_i, A_i)$,
- (b) $\forall F \in \mathcal{F}, B_j \in F$ for infinitely many j ,

we have $\psi \in B_0 \leftrightarrow A_0 A_1 \dots \models \psi$

- Conversely, assume $\varphi_1 \cup \varphi_2 \in B_0$. Then $\varphi_2 \in B_0$ or $\varphi_1, \varphi_1 \cup \varphi_2 \in B_0$.
- If $\varphi_2 \in B_0$, by induction hypothesis, $A_0 A_1 \dots \models \varphi_2$, and hence $A_0 A_1 \dots \models \varphi_1 \cup \varphi_2$.
- If $\varphi_1, \varphi_1 \cup \varphi_2 \in B_0$. Assume $\varphi_2 \notin B_j$ for all $j \geq 0$. Then $\varphi_1, \varphi_1 \cup \varphi_2 \in B_j$ for all $j \geq 0$.
- Since $B_0 B_1 \dots$ satisfies (b), $B_j \in F_{\varphi_1 \cup \varphi_2}$ for infinitely many $j \geq 0$, we obtain a contradiction.
- Thus, $\exists$ a smallest k s.t. $\varphi_2 \in B_k$. Then by induction hypothesis, $A_i A_{i+1} \dots \models \varphi_1$ and $A_k A_{k+1} \dots \models \varphi_2$ for all $i < k$.
- Hence, $A_0 A_1 \dots \models \varphi_1 \cup \varphi_2$.

GNBA Size

- ▶ States of A_φ are subsets of $CI(\varphi)$

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- ▶ States of A_φ are subsets of $CI(\varphi)$
- ▶ Maximum number of states $\leq 2^{|\varphi|}$
- ▶ Number of sets in $\mathcal{F} = |\varphi|$
- ▶ LTL $\varphi \rightsquigarrow$ NBA A_φ : Number of states in $A_\varphi \leq |\varphi|.2^{|\varphi|}$

Complexity of LTL Modelchecking

- ▶ Given φ , $A_{\neg\varphi}$ has $\leq 2^{|\varphi|}$ states
 - ▶ $|\varphi|$ =size/length of φ , the number of operators in φ
- ▶ $TS \otimes A_{\neg\varphi}$ has $\leq |TS|.2^{|\varphi|}$ states
- ▶ Persistence checking : Checking $\Box\Diamond\eta$ on $TS \otimes A_{\neg\varphi}$ takes time linear in $|\eta|.|TS \otimes A_{\neg\varphi}|$

A Weak Lower Bound

The hamiltonian path problem is polynomially reducible to the complement of the LTL modelchecking problem.

- ▶ Given graph $G = (V, E)$ synthesize in polynomial time a TS and an LTL formula φ
- ▶ Show that G has a HP iff $TS \not\models \varphi$

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- ▶ Show that G has a HP iff $TS \not\models \varphi$
- ▶ TS is the graph itself, with one new node added, say b such that all vertices of G have an edge to b , and b has a self loop. Let the label of a node in the TS be the name of the vertex.
- ▶ Write an LTL formula to capture absence of a HP in G . Assume $V = \{v_1, \dots, v_n\}$.
- ▶ The formula $\varphi = \neg\psi$ where ψ is

$$(\Diamond v_1 \wedge \Box(v_1 \rightarrow \bigcirc \Box \neg v_1)) \wedge \dots (\Diamond v_n \wedge \Box(v_n \rightarrow \bigcirc \Box \neg v_n))$$

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- ▶ So LTL model checking is co-NP hard as HP is NP-complete.
- ▶ Actual complexity of LTL model checking : PSPACE-complete.
For this, show that given a LBTM M and a word w , construct in poly time a TS and an LTL formula φ such that M accepts w iff $TS \models \varphi$.

Summary

- ▶ Propositional Logic (natural deduction, satisfiability checking, normal forms, soundness, completeness, Horn formulae, resolution, compactness)
- ▶ First Order logic (normal forms, decidability of satisfiability for fragments, general non-termination of resolution, model checking)
- ▶ MSO (decidability of satisfiability for fragments, model checking)
- ▶ Linear Time Temporal Logic (satisfiability and model checking)

Forward

- ▶ $\exists\forall$ Automata and the LTL connection
- ▶ Impossibility of capturing some properties in FO/LTL
- ▶ Equivalence of LTL and FO
- ▶ Automata classes equivalent to FO and LTL
- ▶ Other logics : CTL/Dynamic logics
- ▶ Beyond languages : transformations (automata and logic to aid transformations)
- ▶ Quantitative Automata and Logics

Thankyou

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- ▶ Mayukh Majumdar

Last and Final Call

Write LTL formulae for each of the following.

- (a) Finitely many a 's.
- (b) Infinitely many a 's and finitely many b 's.
- (c) Eventually a and eventually forever $\neg a$.
- (d) There is an a which is never eventually followed by two occurrences of b 's.
- (e) There is atleast one a , and b holds continuously since (and after) the last occurrence of a .